Vocabulary Builder

Full Feature Analysis

A technical & competitive deep-dive

Science-Backed | B2B-Ready | Vietnam-Focused

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Spaced Repetition System (SM-2 Algorithm)

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This is the same algorithm used by Anki, widely considered the gold standard in memory science. The app tracks three per-word variables:

Ease Factor

Starting value

2.5

Starts at 2.5. Multiplied by the review interval after each recall. Increases with strong recall, decreases with poor recall.

Review Interval

Days until next review

Days

Number of days before the word is shown again. Grows after each successful recall.

Review Count

Per word

N+

The number of times a word has been reviewed. Used with accuracy to determine mastery level.

Poor recall resets the interval; strong recall extends it exponentially — optimizing long-term retention while minimizing study time.

Three Mastery Levels & Flashcard Mode

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Learning

Default for new words

Reviewing

Reached after 3+ reviews with "e70% accuracy

Mastered

Reached after 10+ reviews with "e90% accuracy and a "e30-day review interval

FLASHCARD MODE

Front/back card review with audio pronunciation (fetched from the dictionary API) and example sentences. Integrated directly with the SM-2 engine so every flip updates that word's review schedule.

Quiz Mode & Daily Streak Tracking

Quiz Mode

Multiple-choice format. Up to 10 words per session drawn from the user's active list. Shows the definition, user selects the correct word from 4 options. Tracks score and percentage at the end.

Daily Streak Tracking

Day-over-day consistency is measured and displayed. The streak increments for consecutive calendar days of activity and resets if a day is missed — a proven motivational mechanic.

Word & Content Management

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Manual Word Entry

- Word, definition, example sentence, and custom tags — all user-controlled
- Words can be edited or deleted at any time

Search & Filter

Live search across word and definition fields. Filterable by mastery level and tags.

Tag System

Custom tags per word (or per import batch) for organizing vocabulary by topic, unit, or source.

Automatic Dictionary Lookup

When a user adds a word, the app queries a dictionary API (proxied through the backend) to auto-populate definitions, example sentences, phonetic pronunciation, and audio.

The Datamuse API is also queried for synonyms and antonyms.

Users can also manually trigger the "Word Lookup" view to search any word before adding it.

CSV Bulk Import

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Users upload a file with columns: word, definition, sentence. The app:

1. Auto-detects whether a header row is present
2. Recognizes common header variations ("term", "vocab", "meaning", "phrase", etc.)
3. Skips duplicates (already in the user's list)
4. Skips rows missing word or definition
5. Shows a preview with counts (valid, duplicates, invalid) before committing
6. Lets users apply a tag to the entire import batch

This is the key feature for businesses who want to load industry-specific word lists.

Data & Sync

Cloud Storage — Firebase Firestore

Every word and its entire review history is stored in Firebase's real-time NoSQL cloud database, tied to the authenticated user's UID. This means:

- Words are synced in real time across every device the user logs into
- No manual backup needed
- Data persists if the user reinstalls the app or switches phones/computers

Subscriber Registry — PostgreSQL

A separate server-side database (PostgreSQL) maintains subscription status, trial dates, admin grants, and payment records. This is used by the admin panel and survives server restarts and redeployments — unlike file-based storage that some competing apps use.

Authentication & Security

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Authentication

Email + Password signup and login

Password reset via email

Anonymous sign-in for the direct checkout flow (no friction for purchasers)

Session persistence — users stay logged in across browser sessions

Security

- Helmet.js HTTP security headers
- API rate limiting
- Webhook signature verification (Paddle webhooks are validated before processing)
- Firebase Security Rules restrict each user's data to their own UID
- Admin password stored as an environment secret, never hardcoded

Payments & Business Model

PRICING

500,000 «

per 6 months · ≈$19 USD

+ 10% VAT for Vietnamese purchasers.
Total: 550,000 «.

7-Day Free Trial

Standard access flow. Full features from day one.

Direct Subscription

"Skip Trial" flow for businesses and employees who want to pay immediately without a trial.

PADDLE (GLOBAL PAYMENT PROCESSOR)

Handles credit/debit cards, Apple Pay, Google Pay, and PayPal. Paddle manages tax compliance, recurring billing, cancellations, and refunds globally.

VIETQR / TECHCOMBANK

Local Vietnamese bank transfer option. The user enters their email, the QR code is revealed, they transfer via any Vietnamese banking app, and the admin activates the account. This is critical for the Vietnamese B2B market where many businesses still prefer bank transfer over card.

UI & User Experience

Mobile-First Design

Built specifically to feel like a native mobile app in the browser. Vertical navigation, touch-friendly card sizes, 3D button press effects.

Dark Mode

Full dark/light mode toggle with preference saved to the browser. Every color, border, and gradient is theme-aware.

Responsive Layout

Works on phone, tablet, and desktop browsers without a separate mobile app.

No App Store Required

Runs entirely in the browser — no installation, no App Store approval, no update process for the user.

Progressive Loading

Splash screen while Firebase initializes. "Checking access..." screen while subscription loads. Prevents blank/broken states.

Where This App Stands vs. Competitors

Feature	Vocabulary Builder	Anki	Quizlet	Duolingo	Memrise
SM-2 Spaced Repetition	' Full	' Full	'L Basic	'L Gamified	' Partial
CSV Industry Word Import	' Yes	' (complex)	' (paid tier)	'L	'L
Auto Dictionary + Audio	' Yes	'L Manual	' Limited	' Built-in	' Limited
Synonyms / Antonyms	' Yes	'L	'L	'L	'L
Cross-Device Cloud Sync	' Realtime	& þ AnkiWeb (paid)	' Yes	' Yes	' Yes
Works in Browser (no install)	' Yes	'L App required	' Yes	' Yes	' Yes
Dark Mode	' Yes	' Yes	' Yes	'L	'L
Vietnamese Bank Payment	' VietQR	'L	'L	'L	'L
Admin Override / B2B Console	' Full panel	'L	'L	'L	'L
Industry-Specific Focus	' Designed for it	& þ General	& þ General	'L Consumer only	& þ General
One-Time Bulk Team Setup	' CSV batch	& þ Manual	& þ Limited	'L	'L
Price (6 months)	~$19	Free / $25/yr (sync)	$35.99/yr	$79.99/yr	$59.99/yr

The Core Competitive Argument

Against Anki

Anki requires a separate desktop app and AnkiWeb subscription for sync. It has a steep learning curve and no auto-dictionary lookup. Vocabulary Builder offers the same SM-2 algorithm in a clean browser interface with automatic definitions, synonyms, and audio — at a lower cost.

Against Quizlet

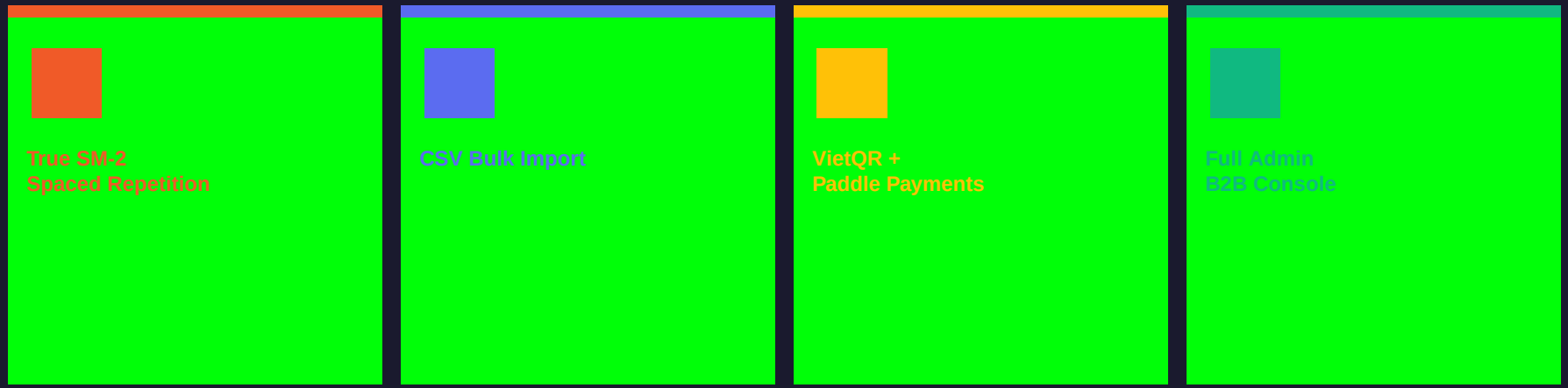
Quizlet's free tier is increasingly limited, and its "spaced repetition" is basic compared to true SM-2. CSV import is only on paid plans. Vocabulary Builder offers deeper SRS, automatic word enrichment, and is purpose-built for professional vocabulary — not just student flashcards.

Against Duolingo / Babbel

Those are designed for complete language beginners following a fixed curriculum. They cannot be customized with your own industry vocabulary. Vocabulary Builder is for professionals who already know conversational English and need to master the specific vocabulary of their field.

Against generic apps in Vietnam

No other app combines SM-2 SRS + industry CSV import + auto-dictionary lookup + VietQR payment + a Vietnamese B2B admin console in one product.

SUMMARY

One App. Every Industry.
Every Word.